

The Wall

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Independent

Abstract

What it is to stand at the wall.

The wall is not a failure.

You arrive at it because you were precise. You eliminated everything else. You narrowed the problem to its exact point and then you stood at the point and it did not give.

A wall means you got close. Most work never gets close. Most work stays in the middle distance, describing the countryside, estimating the height of things. To reach the wall you have to have crossed all that. You have to have known what the problem actually was.

The wall is different from not knowing. Not knowing is diffuse. The wall is sharp. You know exactly what you cannot do. You can say it in one sentence.

For the Riemann Hypothesis the sentence is: *the non-Toeplitz correction cannot be shown to lie in ℓ^2* . That is the whole problem. One sentence.

For Navier–Stokes: *the cubic enstrophy inequality does not close*. One sentence.

For P versus NP: harder to say in one sentence. This is why P versus NP is different. For P versus NP, the wall is not even reachable. You cannot see it yet. You are still in the middle distance. This is a different kind of difficulty.

The walls we have reached are achievements. They look like failure from outside. From inside they feel like precision.

Standing at the wall: you know exactly where you are. You know what you would need. You do not have it. You stay.

This is the position of mathematics at the millennium. Standing. Knowing. Not having. Staying.